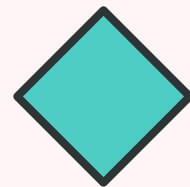


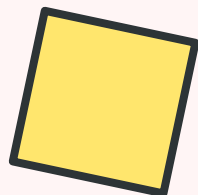
ICLR 2025



Task-Adaptive Pretrained Language Models via Clustered Importance Sampling

David Grangier, Simin Fan, Skyler Seto, Pierre Ablin

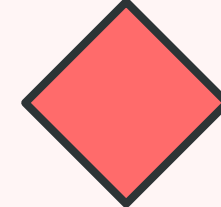
Apple



Data Selection



Presentation Overview



01

Introduction & Motivation

Understanding the challenge of training specialist language models with limited domain data

02

Related Work

Contextualizing CRISP within the landscape of language model training and data selection

03

The CRISP Methodology

Deep dive into the clustered importance sampling approach for task-adaptive pretraining

04

Experimental Setup

Detailed configuration of models, datasets, and evaluation protocols

05

Language Modeling Results

Empirical evaluation of CRISP on domain-specific language modeling tasks

06

MCQ Results

Evaluation on end-tasks with accuracy metrics rather than perplexity

07

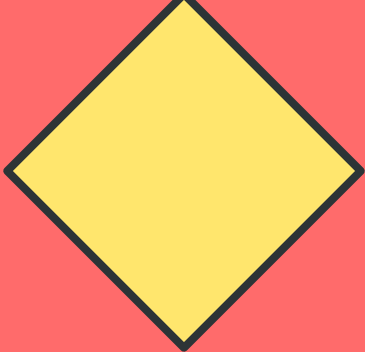
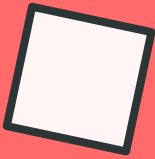
Detailed Analysis

Ablation studies and in-depth examination of CRISP's behavior

08

Conclusions & Future Work

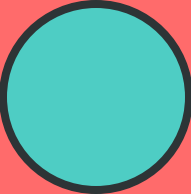
Synthesizing key insights and outlining directions for future research



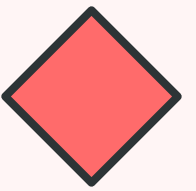
01

Introduction & Motivation

Understanding the challenge of training specialist language models with
limited domain data



The Specialist vs Generalist Dilemma



Generalist Language Models

✓ Strengths

- ▶ Address wide variety of tasks without explicit training
- ▶ Provide good initialization for fine-tuning
- ▶ Compete with dedicated models on many tasks

✗ Limitations

- ▶ Require large training sets (>100B tokens)
- ▶ Need large models to fit comprehensive data
- ▶ High training and inference costs



Specialist Language Models

✓ Strengths

- ▶ Better accuracy at same model size
- ▶ Lower inference costs
- ▶ Focused on specific domain/task

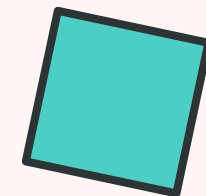
✗ Limitations

- ▶ Limited to narrow specialty
- ▶ Require large specialist datasets
- ▶ Specialist data often unavailable



Key Example: 175B parameter GPT-3 is **less accurate** than 136M-parameter specialist on English-to-German translation

The Core Problem: Limited Specialist Data



Pretraining Requirements

100B+

Tokens Required

- Modern LMs commonly pre-trained on datasets larger than 100B tokens
- Equivalent to ~1 million books or 60× annual publication of top English publisher
- This amount cannot be commissioned for most applications



This is the limiting factor preventing specialist model pretraining for most domains

When is Specialist Pretraining Interesting?



Condition (i)

The targeted task justifies the cost of training a dedicated model
→ Dependent on application and economic benefit



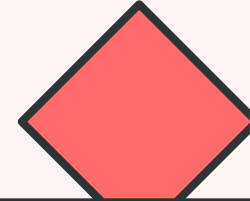
Condition (ii)

A specialist dataset large enough for pretraining is available
→ More limiting - 100B+ tokens rarely available

Our Focus

Relaxing condition (ii) by studying methods to train specialist models when specialized data is scarce

The Proposed Solution: CRISP



ClusteRed Importance SamPling

A novel method for task-adaptive pretraining of language models

Core Mechanism

1

Cluster

Cluster the generalist dataset into meaningful groups

2

Analyze

Compute cluster frequencies in the specialist dataset

3

Sample

Resample generalist data matching specialist distribution

Key Properties



Scalable

Works with millions of clusters and billions of tokens



Flexible

Suitable for both pretraining and continued pretraining



Multi-task

Works well in multi-task settings with averaged weights



Effective

Systematically outperforms alternative methods



Given: Large generalist dataset (Dg) + Small specialist dataset (Ds) → **Train:** Specialist model with modified distribution guided by Ds

Key Contributions & Findings

1

Novel Method

Introduce clustered importance sampling for task-adaptive pretraining, building upon recent work but scaling to millions of clusters

Innovation

First to show IS works for end-tasks like MCQ

2

Exhaustive Evaluation

Comprehensive comparison across model sizes and cluster configurations

350M - 7B

Parameters

64 - 16M

Clusters

3

Strong Results

Demonstrate improvements across different domains in both perplexity and accuracy metrics

✓ Language modeling perplexity gains

✓ MCQ accuracy improvements

4

Deep Analysis

Ablation studies examining impact of dataset sizes, clustering configurations, and model sizes

► Clustering representation impact

► Optimal cluster count analysis

► Multitask and continued pretraining

3 LM Tasks

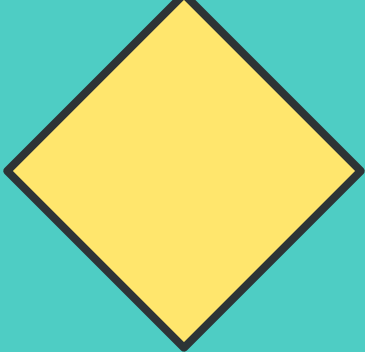
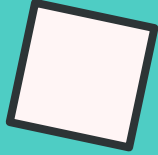
Medical, Programming, Encyclopedia

3 MCQ Tasks

ARC, MMLU, RWDB-R

30× Speedup

Training cost reduction



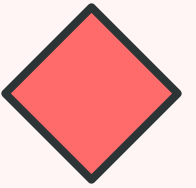
02

Related Work

Contextualizing CRISP within the landscape of language model training and data selection



Generalist vs Specialist Language Models



Generalist Language Models

Capabilities

- ▶ Address tasks without explicit training (few-shot)
- ▶ Good initialization for fine-tuning
- ▶ Compete with dedicated models (Jiang et al., 2024; Dubey et al., 2024)

Costs

- ▶ Must be much larger than specialists for same accuracy
- ▶ Parameter count directly impacts training/inference costs
- ▶ Require diverse, comprehensive training data

Specialist Language Models

Advantages

- ▶ Better accuracy at given model size (Kerner, 2024)
- ▶ Lower inference costs
- ▶ Focused expertise in narrow domain

Requirements

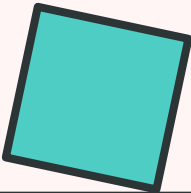
- ▶ Large specialist corpora (billions of tokens)
- ▶ Pretraining or continued pretraining
- ▶ Limited applicability without sufficient data



Trade-off: Generality vs Efficiency

Generalists sacrifice efficiency for breadth; specialists sacrifice breadth for efficiency. The challenge is achieving specialist efficiency when specialist data is scarce.

Existing Specialist Model Approaches



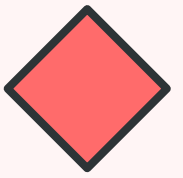
Medicine & Biology	Programming & Math	Finance
BioBERT / PubMedBERT Lewis et al., 2020 BioMistral Labrak et al., 2024 BiomedLM Bolton et al., 2024	Minerva Lewkowycz et al., 2022 Code Llama Rozière et al., 2024 Llemma Azerbayev et al., 2024	BloombergGPT Wu et al., 2023 Pixiu Xie et al., 2023a ... Other domain-specific models
Data Available Billions of tokens	Data Available Billions of tokens	Data Available Billions of tokens

Common Pattern



All successful specialist models require billions of tokens for pretraining or continued pretraining. When specialist data is limited, task-adaptive data-selection methods train specialist models on generalist data instead.

Task-Adaptive Data Selection Methods



1

Vocabulary & Overlap Methods

Vocabulary Overlap

Gururangan et al. (2020): Continued pretraining improves when using data with high vocabulary overlap with targeted task

Nearest Neighbors

Aharoni & Goldberg (2020): Select data using nearest neighbors of specialist set with BERT sentence distance

2

Classifier-Based Methods

LM Score Contrast

Moore & Lewis (2010), Axelrod et al. (2011): Contrast generalist vs specialist LM scores

Logistic Regression

Iter & Grangier (2021): Train classifiers to identify in-domain data

Fine-tuned BERT

Various: Use BERT-based classifiers for domain identification

3

Importance Sampling & Gradient Methods

Importance Sampling

Xie et al. (2023c): Select pretraining data with similar distribution using hashed-ngrams. Grangier et al. (2024b): Cluster-based IS (foundation for CRISP)

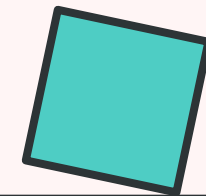
Gradient Alignment

Pruthi et al. (2020), Xia et al. (2024), Grangier et al. (2023): Bilevel optimization to align gradients from generalist and specialist data

...

Other approaches: Reinforcement learning (Liu et al., 2019; Yoon et al., 2020), Bayesian optimization (Ruder & Plank, 2017), Data models (Ilyas et al., 2022), Influence models (Yu et al., 2024)

Pretraining Data Quality Considerations



Data Quality Dimensions

Language Filtering

Remove data in other languages (Cook & Lui, 2012)

Text Formatting

Clean and standardize text structure (Xu et al., 2024)

Content Length

Favor long-form text (Gao et al., 2021; Gunasekar et al., 2023)

Deduplication

Remove duplicate content (Lee et al., 2022)

Source Balancing

Balance different data sources (Xie et al., 2023b; Fan et al., 2024)

Quality Filtering

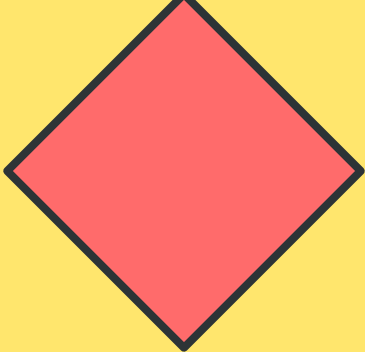
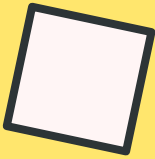
Filter, correct, or generate data with LMs (Kong et al., 2024; Chen & Mueller, 2024; Maini et al., 2024)

Orthogonal to Domain Adaptation

Data quality considerations are applied alongside domain adaptation decisions. Quality filters and domain selection are complementary processes that together determine the final training set composition.



Key Insight: Quality filtering improves the base pretraining loss, while domain adaptation focuses that quality on the target task. Both are essential for optimal results.



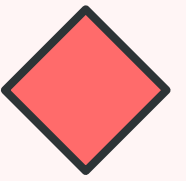
03

The CRISP Methodology

Deep dive into the clustered importance sampling approach for task-
adaptive pretraining



Problem Formulation & Notations



Dataset Definitions

D_g Generalist Dataset

Training dataset sampled from generalist distribution D_g

Example: RedPajama-v2 with 30T+ tokens

D_s Specialist Dataset

Dataset representative of final task, sampled from specialist distribution $D_s \neq D_g$

Example: Medical, programming, or exam questions

Loss Function & Perplexity

Loss Definition

$$\begin{aligned} L(D; \theta) &= (1/|D|) \sum \ell(x; \theta) \\ &= -(1/|D|) \sum (1/|x|) \sum \log p(x_{i|1:i-1}; \theta) \end{aligned}$$

Average negative log-likelihood over dataset D

Perplexity Definition

$$P(D; \theta) = \exp(L(D; \theta))$$

Exponential of loss - measures model's predictive uncertainty



Objective

Train a model θ that minimizes $L(D_s; \theta)$ using guidance from small D_s to sample from large D_g

Method 1: Classification Approach

Core Concept

Train a binary classifier to estimate the probability that a generalist pretraining document belongs to the targeted domain

$$C(D_g, t) = \{x \in D_g : \phi(x) > t\}$$

Where t is the classification threshold

Training Process

1

Positive Examples

Sample from specialist dataset D_s

2

Negative Examples

Sample subset from generalist dataset D_g

3

Classifier Training

Learn ϕ to distinguish domain membership

Implementation Details

Classifier Architecture

- ▶ Logistic regression classifier
- ▶ Trained over SBERT embeddings
- ▶ Established method (Minaee et al., 2021)

Threshold Selection

- ▶ Sensitive hyperparameter
- ▶ Trade-off: focus vs dataset size
- ▶ 2.5% threshold worked best in experiments

Note: SBERT representation also used in alternative selection strategies for fair comparison

Method 2: Gradient-Alignment (DoGE)



Core Concept

Select weights for different data sources by aligning gradients from generalist and specialist datasets using bilevel optimization

Pretraining Loss

$$L((w, Dg); \theta) = \sum w_i \cdot L(Dg_i; \theta)$$

Reweighed loss across data sources

Targeted Loss

$$L(Ds; \theta)$$

Loss on specialist dataset

Bilevel Optimization

Inner problem: $\theta^*(w) = \arg \min_{\theta} L((w, Dg); \theta)$

Outer problem: Find w that minimizes $L(Ds; \theta^*(w))$

Algorithms select weights during pretraining to align gradients w.r.t. θ

Implementation & Limitations

DoGE Algorithm

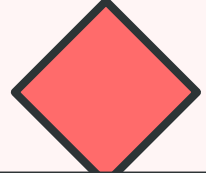
- ▶ Fan et al., 2024
- ▶ Applied over 64 clusters
- ▶ Learning mixture weight of dim 64

Scalability Challenge

- ▶ Harder to scale to large models
- ▶ Find weights with small model first
- ▶ Transfer to larger model

 **Clustering:** Use k-means on SBERT embeddings to identify data clusters (not pre-segmented sources)

Method 3: CRISP - Core Concept



Strategy Overview

1

Divide Space

Divide text space into clusters using embeddings

2

Decompose Loss

Decompose specialist and generalist loss as weighted sum over clusters

3

Independence

Assume loss per cluster is identical regardless of data source



Key Insight

If the loss in a cluster is the same whether its data comes from Dg or Ds, we can compute the specialist loss as the generalist loss with cluster reweighting

Importance Sampling Application

Importance Weights

$$w(c) = P(c|Ds) / P(c|Dg)$$

Ratio of cluster frequencies

Reweighted Loss

$$L(Ds; \theta) = E[w(c) \cdot L(c; \theta)]$$

Sample from generalist, weight by specialist

Practical Estimation

P(c|Ds) - Specialist

Count cluster frequencies on small Ds

L(c; θ) - Generalist

Estimate from large Dg with less variance

CRISP: Mathematical Foundation



Objective

Identify model with low loss on specialist distribution D_s

Mathematical Derivation

1

Initial Definition

$$L(D_s; \theta) = E_{x \sim D_s}[\ell(x; \theta)] = \sum_x \ell(x; \theta) P(x|D_s)$$

2

Marginalize Over Clusters

$$L(D_s; \theta) = \sum_c \sum_x \ell(x; \theta) P(x|c, D_s) P(c|D_s)$$

3

Independence Assumption

$$P(x|c, D_s) = P(x|c)$$

Cluster distribution is independent of data source

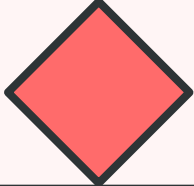
4

Final Form

$$L(D_s; \theta) = \sum_c L(c; \theta) P(c|D_s) = E_{c \sim (c|D_s)}[w(c) \cdot L(c; \theta)]$$

Key Terms Defined

CRISP Algorithm: Step-by-Step



Algorithm 1: CRISP Training

Input: D_s (specialist set), D_g (generalist set)

Params: T (steps), B (batch size)

Step 1: Compute cluster histogram

$h_s \leftarrow \{P(c|D_s), \forall c\}$
Count cluster frequency on specialist set

Step 2: Initialize model

$\theta_0 \leftarrow \text{InitModel}()$

Step 3: Training loop

for $t = 1, \dots, T$ do:
 for $i = 1, \dots, B$ do:
 $c_i \sim \text{Categorical}(h_s)$
 Sample cluster from specialist histogram
 $x_i \sim \text{Uniform}(D_g \cap K(c))$
 Sample example from generalist in cluster
 $\theta_t \leftarrow \text{AdamUpdate}(\theta_{t-1}, \{x_1, \dots, x_B\})$

Key Algorithmic Advantages

 Implicit Reweighting

Sample clusters from importance instead of explicitly reweighing loss

 Reduced Variance

Avoid frequently visiting low-weight clusters, helping convergence

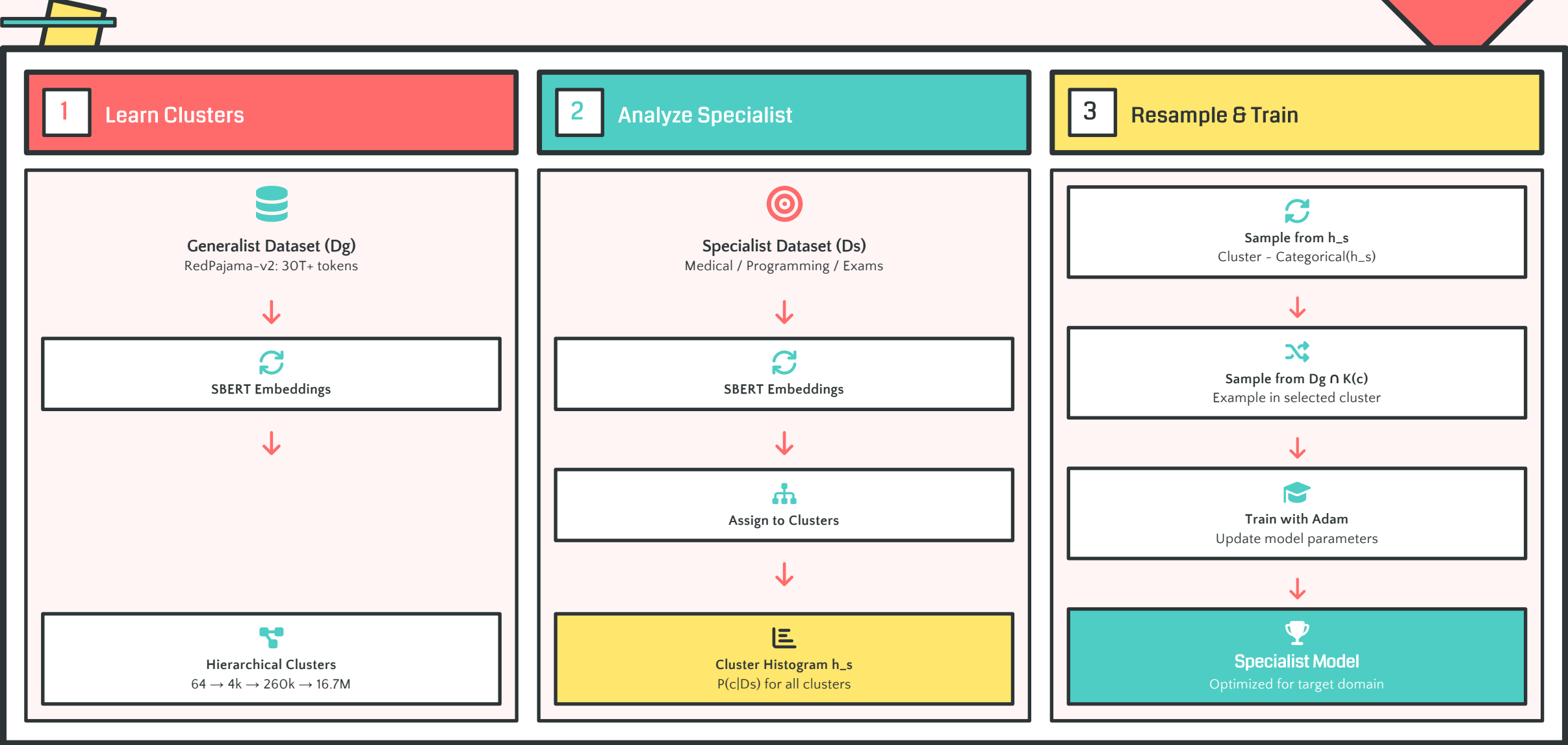
 Efficient

Group generalist examples by cluster prior to training

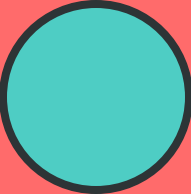
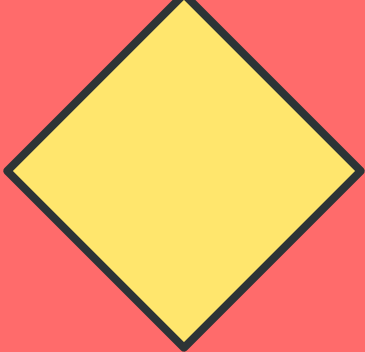
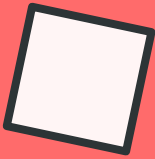
 Simple

Straightforward implementation with standard optimizers

CRISP Workflow Visualization



Generalist Data 30T+ tokens → Specialist Guidance Small Ds → Task-Adaptive Model Best of both worlds

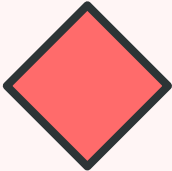


04

Experimental Setup

Detailed configuration of models, datasets, and evaluation protocols

Model Architectures & Sizes



Transformer LM Configurations

Parameter	350M	1.3B	6.7B
Architecture	Transformer (Vaswani et al., 2017)		
Embedding Dim	1,024	2,048	4,096
Latent Dim	4,096	8,192	16,384
Num Heads	16	16	32
Depth	24	24	32
Context Limit	1,024 tokens		
Training Tokens	40B	120B	350B

Optimization Configuration



Optimizer
Adam (Kingma & Ba, 2015)



Batch Size
115k / 96k / 1.04M tokens



Learning Rate
1e-4 / 1e-4 / 3e-4



Grad Clipping
5.0 / 5.0 / 0.1

Generalist Dataset: RedPajama-v2

Dataset Characteristics

 **Scale**
30T+
Total Tokens

 **Tokenizer**
32k
BPE Vocabulary

 **Content**
Web-only
No task-specific additions

Why RedPajama-v2?

 **Web-crawled only**
No interventions to help evaluation tasks

 **No contamination**
No books, encyclopedias, or academic articles added

 **English-focused**
Head+middle English version with high density

 **Openly available**
Together AI Team, 2023

Data Processing Pipeline

1

Segmentation
1,024 token windows (context limit)

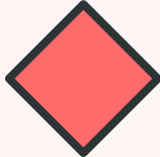
2

Embedding
SBERT for every window

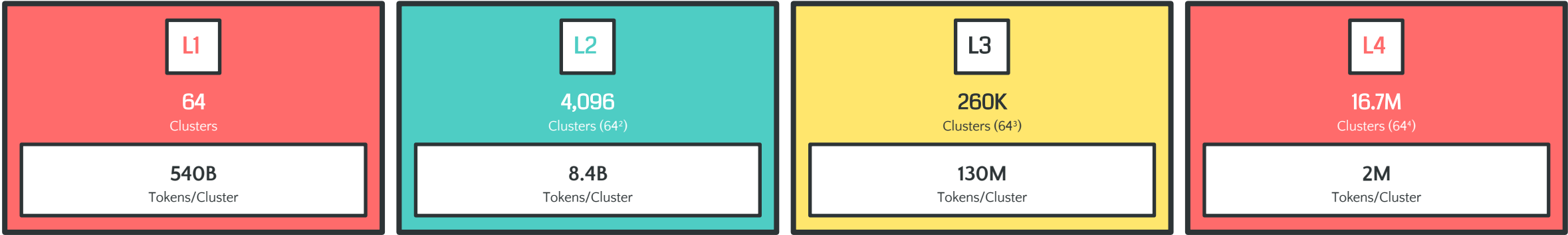
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Clustering
Hierarchical tree structure

Clustering Configuration Details



Hierarchical Clustering Tree



Tree Properties

Structure

- ▶ Tree depth: 4 levels
- ▶ Branching factor: 64
- ▶ k-means++ initialization

Balancing

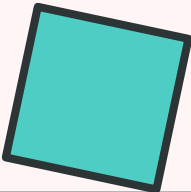
- ▶ Balancing limit: 0.022 ($1.5 \times 1/64$)
- ▶ Split oversized clusters
- ▶ Ensure uniform distribution

Alternative: LSI Embeddings

Also consider Latent Semantic Index (LSI): singular value decomposition over tf-idf representations (Deerwester et al., 1990; Dumais, 2004) as alternative to SBERT

Tree Depth	Branching	Max Clusters
4	64	16.7M

Specialist Tasks: Language Modeling



 PubMed Central

Domain

Medical & Biomedical

Training Tokens

24.0B

Test Tokens

10.3B

From Pile dataset (Gao et al., 2021)

 StackExchange

Domain

Programming Q&A

Training Tokens

359M

Test Tokens

4.68B

From Pile dataset (Gao et al., 2021)

 Wikipedia

Domain

Encyclopedic

Training Tokens

26.7B

Test Tokens

14.1B

From Pile dataset (Gao et al., 2021)

Experimental Setup

Specialist Data Range

14M tokens → Full subset (up to 26.7B)

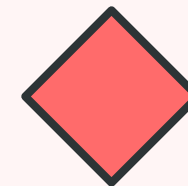
Baselines

Only-specific, Fine-tuned Redpj2

Evaluation Metric

Perplexity (lower is better)

Specialist Tasks: Multiple Choice Questions



ARC

AI2 Reasoning Challenge

Focus
Science questions

Train Tokens
426K

Test Questions
1.18K – 1.19K

4 choices/question



MMLU

Massive Multitask Language Understanding

Focus
Interdisciplinary knowledge

Train Tokens
2.05M

Test Questions
7.09K

4 choices/question



RWDB-R

Reward Bench Reasoning

Focus
Math & programming problems

Train Tokens
144K

Test Questions
695

2 choices/question

Key Differences from LM Tasks



Data Size

Much smaller: 200K – 2M tokens vs 14M – 26.7B



Metric

Accuracy (not perplexity) – model scores answer choices



Loss-Metric Gap

Training loss not closely tied to test metric



Evaluation: Split each dataset into equal train/test sets uniformly at random to avoid cross-contamination between tasks

Unified Experimental Settings

Method-Specific Hyperparameters

Classifier

Threshold

2.5%

Accepts 2.5% of Dg

Regularization

{None, 1000, ..., 0.001}

DoGE

Num Clusters

64

Mixture weight dim

Proxy Model

110M parameters

CRISP

Num Clusters

260K

Main experiments

Ablations

{64, 4K, 260K, 16.7M}

Fair Comparison Setup

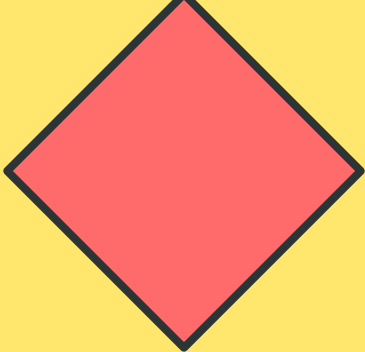
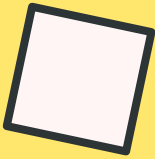
Equal Access

All methods have access to 14M specialist training tokens for fair comparison

Same Hyperparameters

Same optimization settings for all models (1.3B parameters, 120B tokens)

✓ **Validation:** 2.5% threshold worked best for 1.3B models over 120B tokens across multiple validation runs



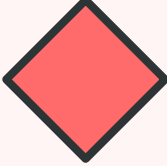
05

Language Modeling Results

Empirical evaluation of CRISP on domain-specific language modeling tasks



Pretraining Results Overview



Key Finding

In-domain perplexity is better with task-dependent pretraining than generic pretraining for all methods

Why Task-Dependent Pretraining Works



Focused Training

Model training focuses on data close to the targeted domain



Efficient Capacity

Model capacity is not used to fit filtered out training data

Specialization Evidence: CRISP on PubMed

97.3%

Better perplexity on PubMed specialist data

95.9%

Worse perplexity on Redpj2 generalist data

This confirms the model specializes: better on target domain, worse on general data

Methods Evaluated

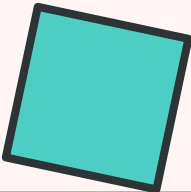
Base Redpj2
Generic pretraining

DoGE
Gradient alignment

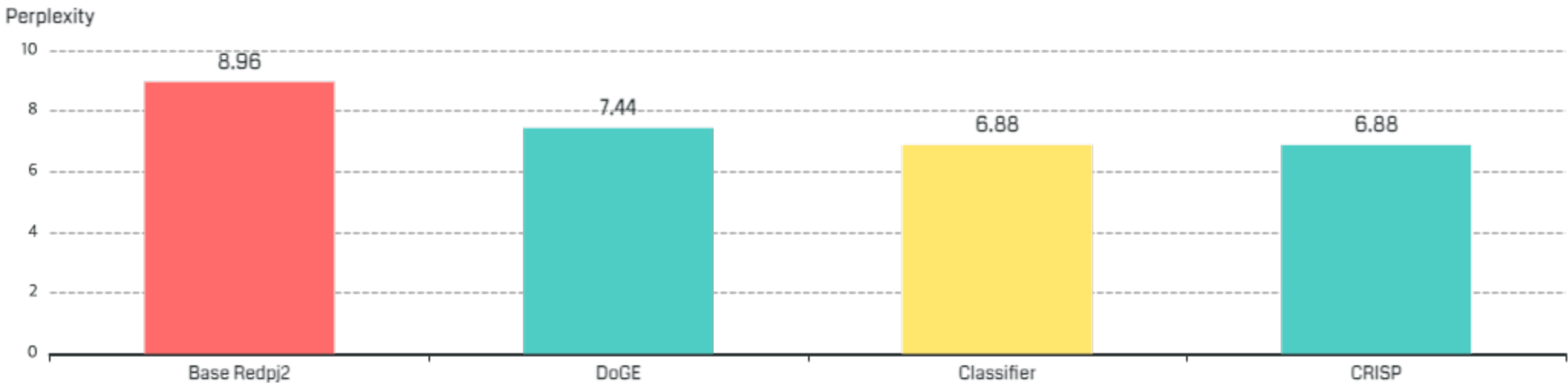
Classifier
Logistic regression

CRISP
Importance sampling

PubMed Pretraining Results



Perplexity Comparison (Lower is Better)



Base Redpj2

8.96

PPL

DoGE

7.44

PPL

Classifier

6.88

PPL

CRISP

6.88

PPL ✓

Improvement over Base

23.2%

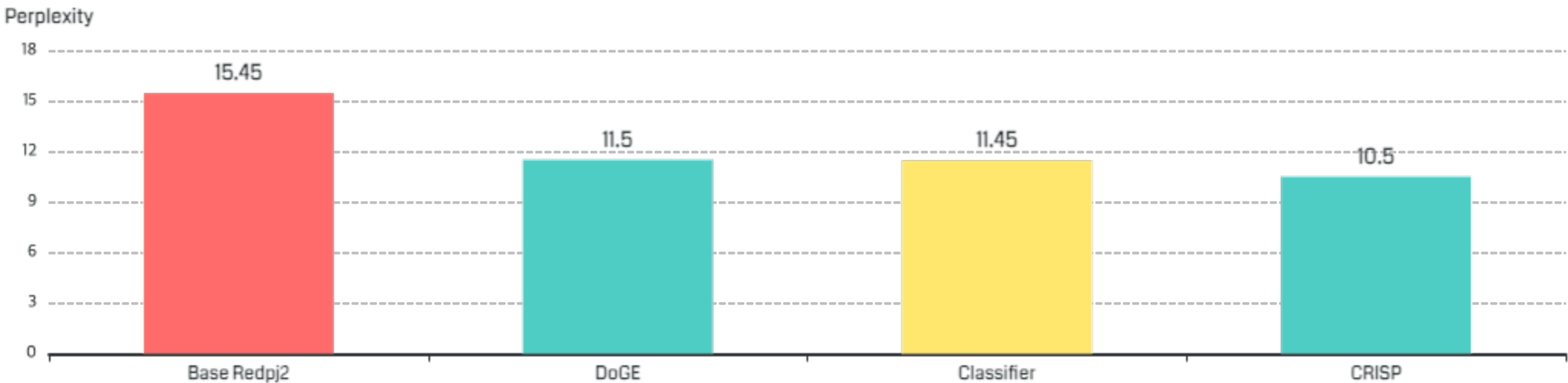
Improvement over DoGE

7.5%

StackExchange Pretraining Results



Perplexity Comparison (Lower is Better)



Base Redpj2

15.45

PPL

DoGE

11.50

PPL

Classifier

11.45

PPL

CRISP

10.50

PPL ✓

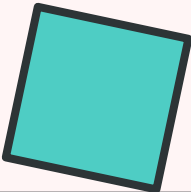
Improvement over Base

32.0%

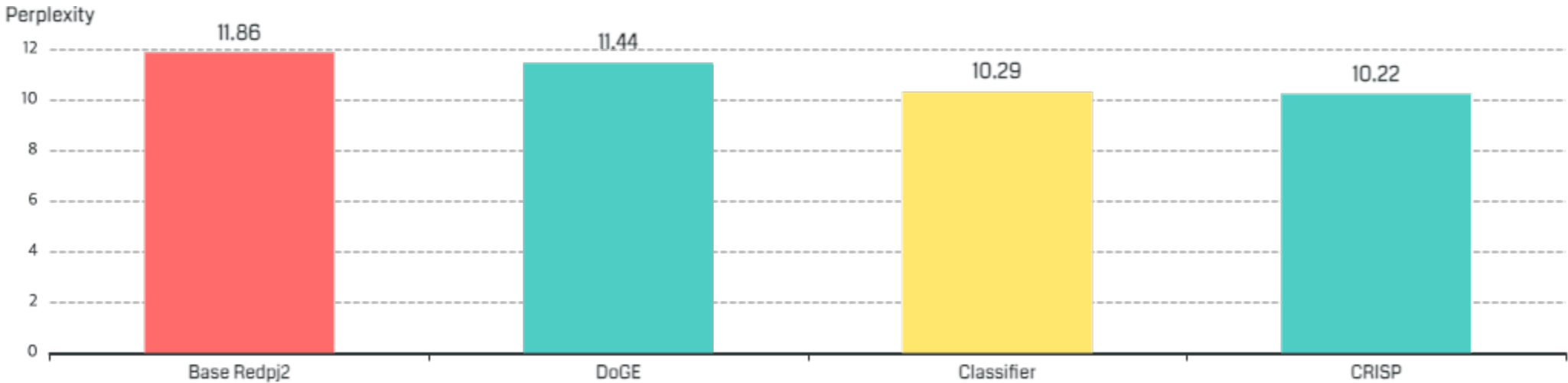
Improvement over DoGE

8.7%

Wikipedia Pretraining Results



Perplexity Comparison (Lower is Better)



Base Redpj2

11.86

PPL

DoGE

11.44

PPL

Classifier

10.29

PPL

CRISP

10.22

PPL ✓

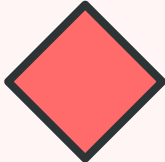
Improvement over Base

13.8%

Improvement over DoGE

10.7%

Fine-Tuning Results Overview



Key Finding

When pretrained models are fine-tuned, the advantage of task-dependent pretraining is preserved

Complementarity: Pretraining + Fine-Tuning



Better Starting Point

Task-specific pretraining checkpoints are better starting points for fine-tuning than generic ones



Complementary Benefits

Shows complementarity between task-dependent pretraining and fine-tuning – both contribute

Necessity of Pretraining



Critical Finding

Below 7B tokens, the "only specific" 1.3B model shows high perplexity. Pretraining on generalist data is essential before domain-specific fine-tuning.

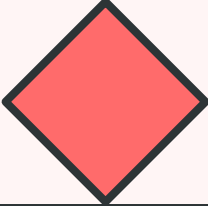
Fine-Tuning Configuration

Fraction Range
3% to 100% specialist

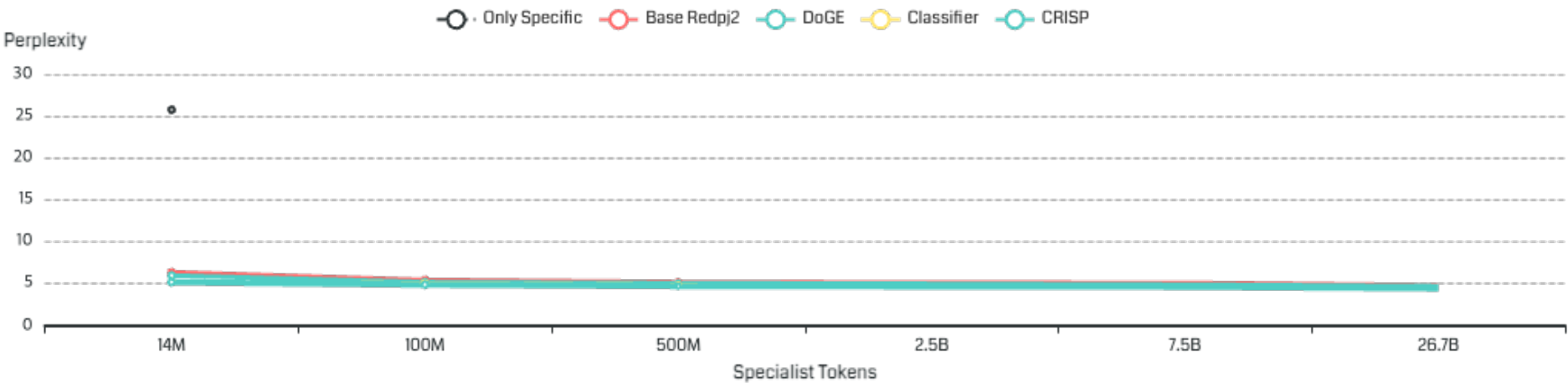
Token Range
14M to 26.7B tokens

Same Models
All pretrained variants

Fine-Tuning: PubMed Results



Fine-Tuned Perplexity vs Specialist Tokens



Key Observations

✓ **CRISP Maintains Lead**
Consistently outperforms other methods after fine-tuning

⬇️ **Ordering Preserved**
Method ranking unchanged from pretraining

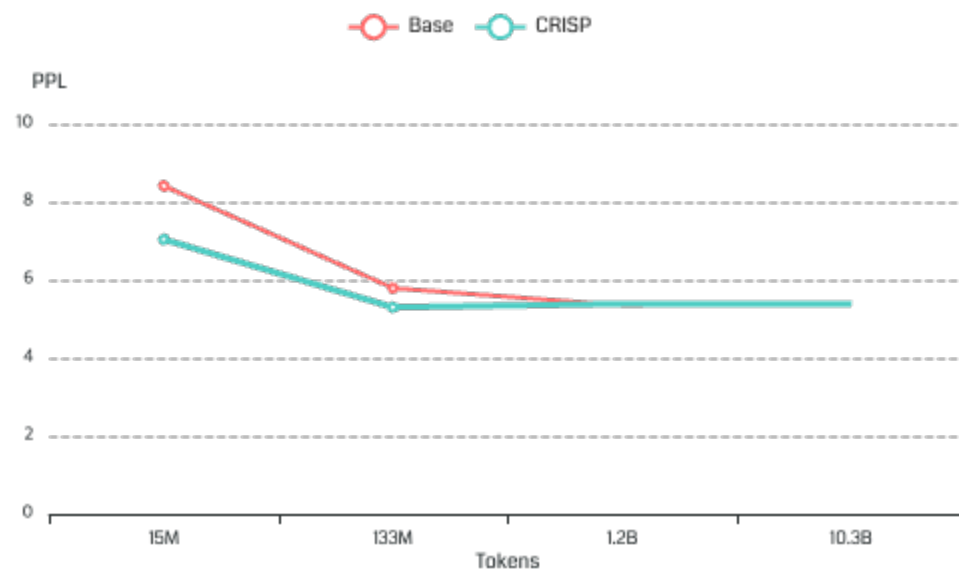
⬇️ **Diminishing Returns**
All methods converge with more specialist tokens

★ **Best at Low Data**
CRISP advantage strongest with limited specialist data

Best PPL (14M) | Best PPL (26.7B) | CRISP at 7.5B
4.20 | 4.44 | 4.63

Fine-Tuning: StackExchange & Wikipedia

StackExchange Fine-Tuning



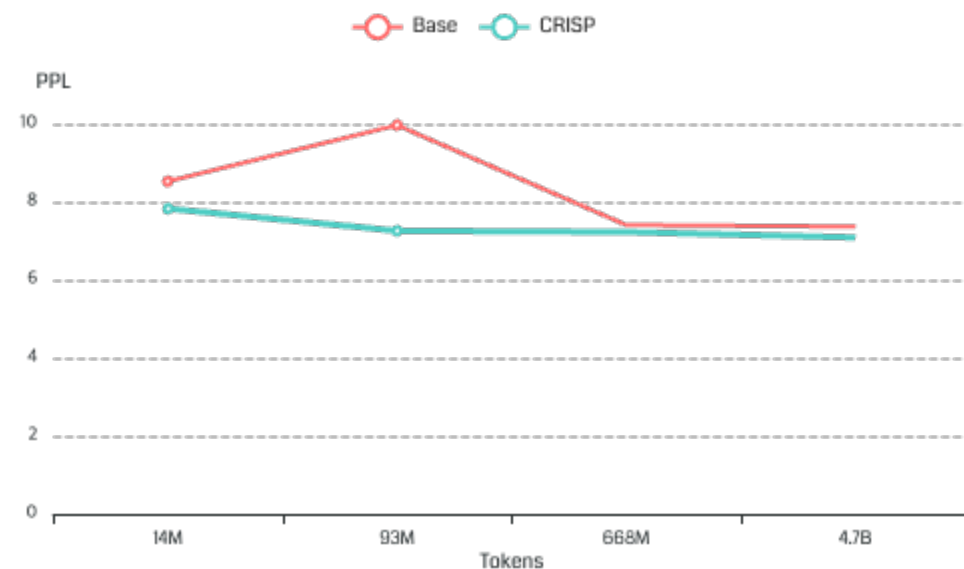
CRISP Best

6.12

Base Best

6.66

Wikipedia Fine-Tuning



CRISP Best

6.40

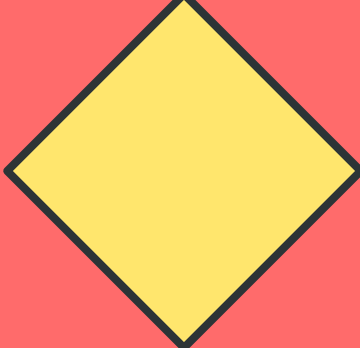
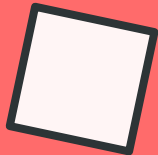
Base Best

7.09



Consistent Pattern Across Domains

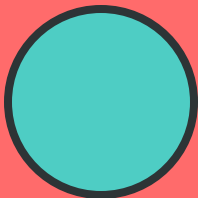
Importance sampling performs better after fine-tuning, even when pretraining results are close. The ordering of methods remains unchanged from pretraining.



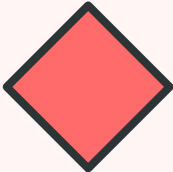
06

Multiple Choice Question Results

Evaluation on end-tasks with accuracy metrics rather than perplexity



MCQ Task Characteristics



Key Differences from LM Tasks

 Data Size

200K-2M

Specialist tokens vs 14M-26.7B for LM

 Metric

Accuracy

Not perplexity - % correct answers

 Loss Gap

Disconnected

Training loss not tied to test metric

MCQ Evaluation Method

 Scoring

Model scores concatenation of question + each possible answer

 Accuracy

Correct when highest score assigned to correct answer

Scoring Variants

Log Probabilities

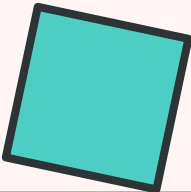
ARC-Easy, RWDB-R

Normalized Scores

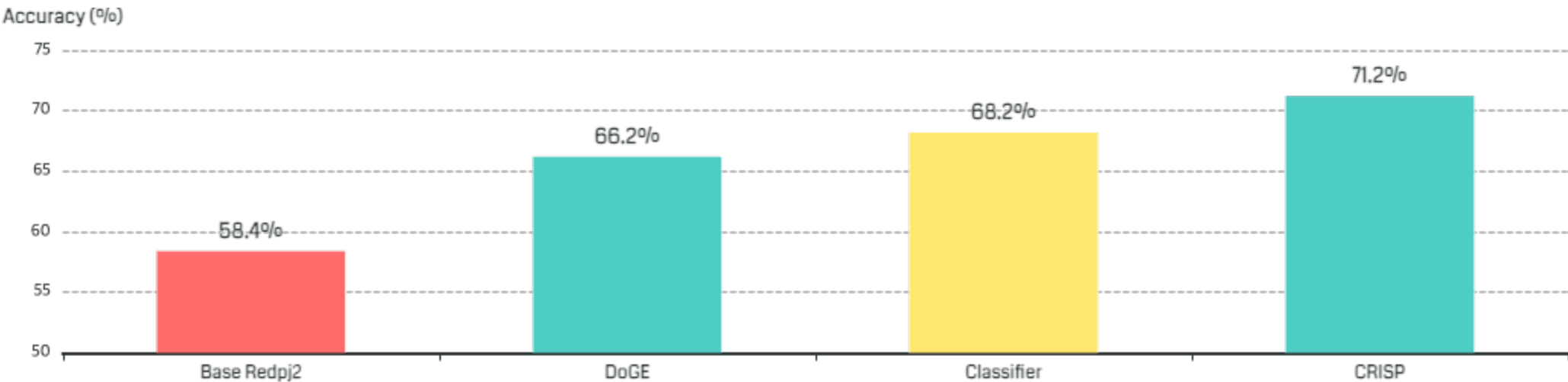
ARC-Challenge, MMLU (log prob / num chars)

 Despite differences: Similar benefits observed for task-dependent pretraining compared to task-agnostic pretraining

ARC-Easy Results



Accuracy Comparison (Higher is Better)



Base Redpj2

58.4%

ACC

DoGE

66.2%

ACC

Classifier

68.2%

ACC

CRISP

71.2%

ACC ✓

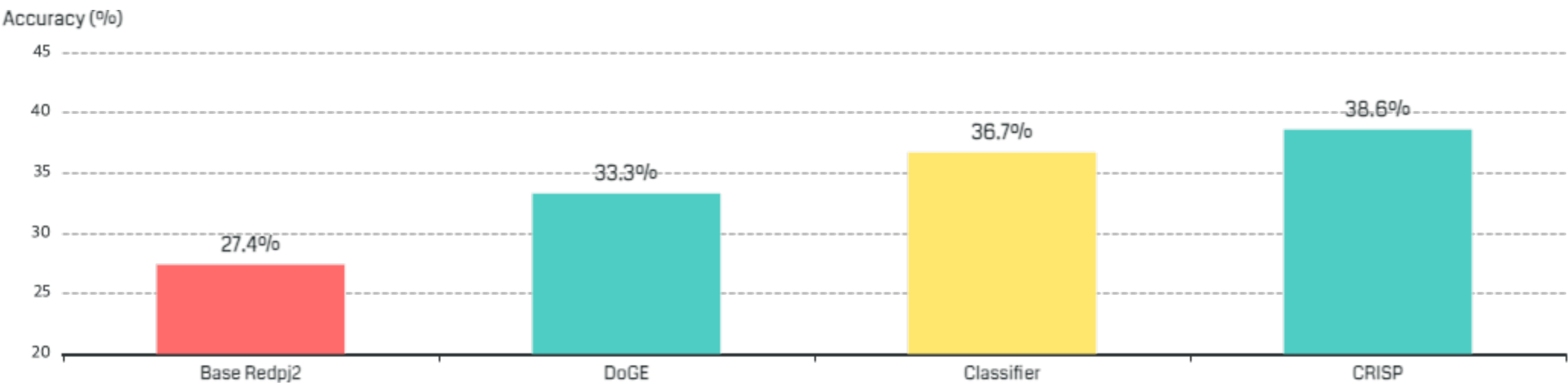
Improvement over Base
+12.8 pts

Improvement over DoGE
+5.0 pts

ARC-Challenge Results



Accuracy Comparison (Higher is Better)



Base Redpj2

27.4%

ACC

DoGE

33.3%

ACC

Classifier

36.7%

ACC

CRISP

38.6%

ACC ✓

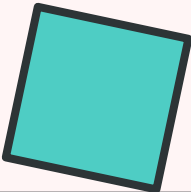
Improvement over Base

+11.2 pts

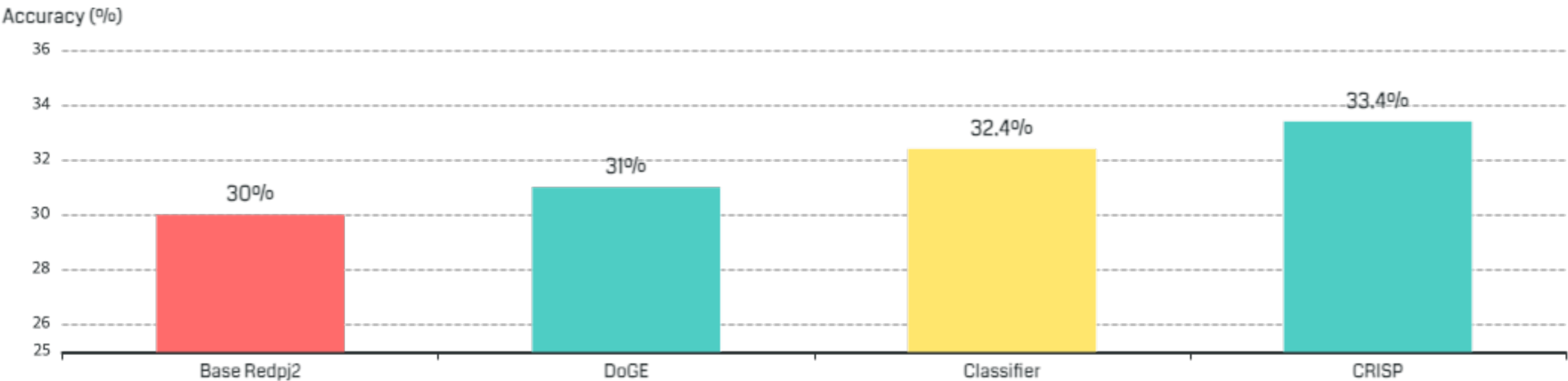
Improvement over DoGE

+5.3 pts

MMLU Results



Accuracy Comparison (Higher is Better)



Base Redpj2

30.0%

ACC

DoGE

31.0%

ACC

Classifier

32.4%

ACC

CRISP

33.4%

ACC ✓

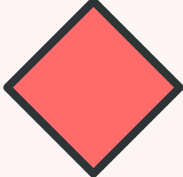
Improvement over Base

+3.4 pts

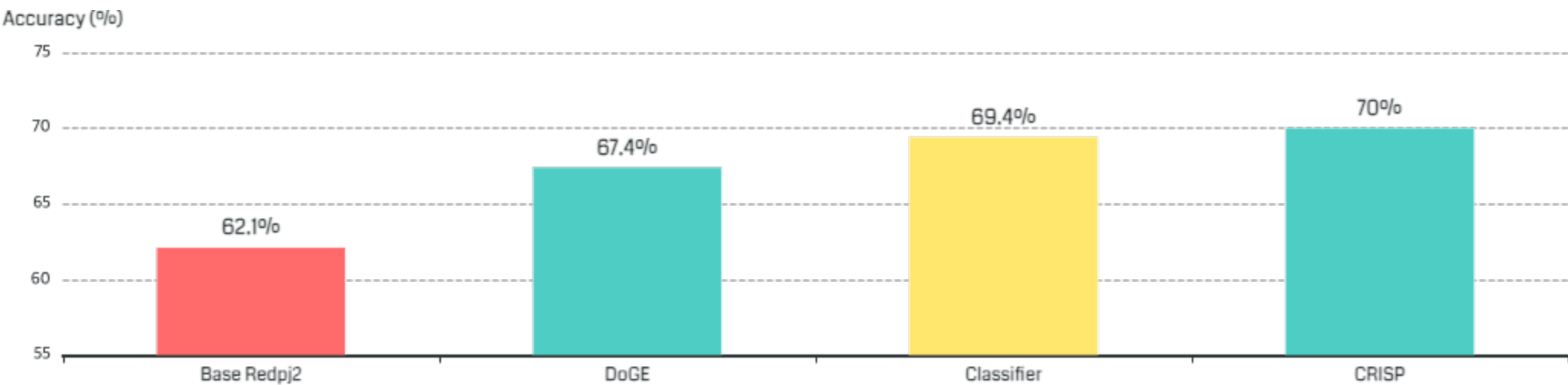
Improvement over DoGE

+2.4 pts

RWDB-R Results



Accuracy Comparison (Higher is Better)



Base Redpj2

62.1%

ACC

DoGE

67.4%

ACC

Classifier

69.4%

ACC

CRISP

70.0%

ACC ✓

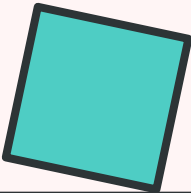
Improvement over Base

+7.9 pts

Improvement over DoGE

+2.6 pts

Fine-Tuning Impact on MCQ Tasks



Key Difference from LM Tasks

Limited benefits from fine-tuning observed on MCQ tasks

Pretraining vs Fine-Tuning Results

ARC-Easy		ARC-Challenge		MMLU		RWDB-R	
Base Pretr:	58.4%	Base Pretr:	27.4%	Base Pretr:	30.0%	Base Pretr:	62.1%
+ Fine-tune:	—	+ Fine-tune:	28.8%	+ Fine-tune:	31.0%	+ Fine-tune:	63.5%
CRISP:	71.2%	CRISP:	38.6%	CRISP:	33.4%	CRISP:	70.0%

Interpretation



Fine-tuning Helps Base

Fine-tuning improves base method on all datasets except ARC-Easy

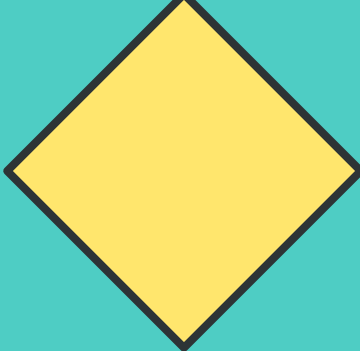
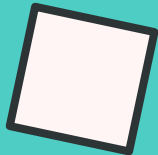


Pretraining Dominates

Not enough to outperform task-specific pretraining – pretraining is more critical for MCQ



Conclusion: Task-dependent pretraining is more critical for MCQ tasks than fine-tuning. The model's initialization from task-adaptive pretraining provides stronger benefits than post-hoc fine-tuning.



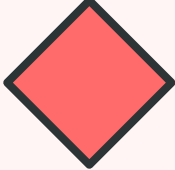
07

Detailed Analysis

Ablation studies and in-depth examination of CRISP's behavior



Clustering: Text Representation Impact



SBERT vs LSI Embeddings

★ SBERT

Type

Sentence-BERT embeddings

Model

MiniLM-L6-v2 (384 dim)

Result

Better in all settings ✓

🧮 LSI

Type

Latent Semantic Indexing

Model

SVD over tf-idf (256 dim)

Result

Lower accuracy

Why SBERT Performs Better

The representation is important because examples in the same cluster should be close in the embedding space

Independence Assumption: $L(D_g \cap K(c); \theta) \approx L(D_s \cap K(c); \theta)$

SBERT better captures semantic similarity, making the independence assumption more valid

💡 **Practical Implication:** Use SBERT embeddings for clustering. The embedding quality directly impacts CRISP's effectiveness.

Clustering: Number of Clusters Trade-off

Accuracy vs Number of Clusters (MMLU)



Trade-off Factors

↓ Fewer Clusters

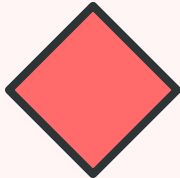
- + Larger clusters
- Stronger hypothesis assumption
- At limit (1 cluster): assumes identical loss everywhere

↑ More Clusters

- + Finer granularity
- Less accurate $P(c|Ds)$ estimation
- Risk of overfitting to training set

✓ **Optimal:** 260K clusters provides best balance for most experiments

Training Data Repetition Analysis



Number of Occurrences per Training Example (MMLU)



1.95

Average Repetitions (16.7M)

1.04

Average Repetitions (260K)

1.00

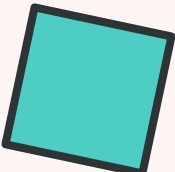
Average Repetitions (4K)

Interpretation

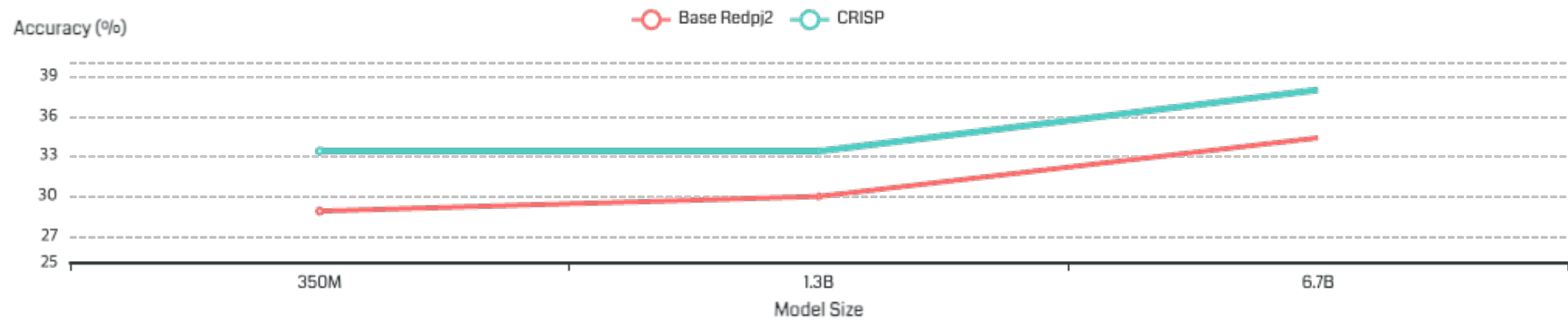
Even for 16.7M clusters, only a small minority of training examples are repeated beyond 10 times. Average repetitions (1.95) well within commonly recommended values (Muennighoff et al., 2023a; Xue et al., 2023).

 Exact repetitions don't fully account for 16.7M cluster performance – training set diversity also matters

Model Size Analysis



Accuracy Across Model Sizes (MMLU)



Consistent Benefit

Benefit of task-dependent training is consistent across model sizes

Training Cost vs Accuracy (ARC-Easy)

1.3B CRISP vs 6.7B Base

~30×

Training Speedup

1.3B CRISP vs 6.7B Base

71.2%

vs 58.4% (ARC-E)

Inference Benefit

5×

Smaller model



Key Insight: A smaller specialist model with CRISP can outperform a much larger generalist model, providing both training and inference efficiency

Task Transfer & Multitasking



Cross-Task Evaluation (1.3B Model)

Pretraining Target	ARC-E	ARC-C	MMLU	RWDB-R
Base Redpj2	58.4	27.5	30.1	62.2
CRISP (ARC)	71.3	38.6	28.9	60.9
CRISP (MMLU)	63.4	28.7	33.4	65.2
CRISP (RWDB-R)	42.4	23.4	26.4	70.1
CRISP (Multi)	68.6	34.1	31.1	70.9

Cross-Task Insights

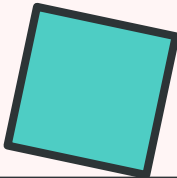
- ▶ Best results when pretraining for target task
- ▶ Transfer differs across tasks
- ▶ MMLU targeting helps all tasks

Multitask Result

- ▶ Best averaged multitask accuracy
- ▶ Equal weight (1/3 each)
- ▶ RWDB-R: multitask > targeted!

💡 Multitask weighting provides flexible, cost-effective solution when targeting multiple related tasks

Continued Pretraining Analysis



Cost-Effective Alternative

Two-Phase Training

Phase 1: Generic
Standard pretraining on Redpj2

Phase 2: Task-Dependent
CRISP continued pretraining

Cost Sharing

Generic pretraining cost can be shared across multiple tasks – train once, specialize many times

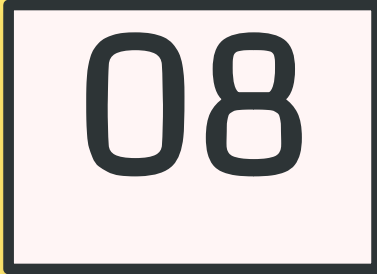
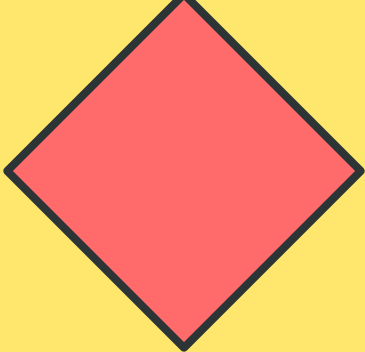
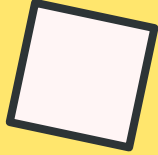
MMLU Continued Pretraining Results



Key Finding

Even 10% CRISP continued pretraining (928 steps out of 1,024) gives accuracy (32.9%) close to full CRISP run (33.4%). Impact stronger than fine-tuning generic model on MMLU (31.0%).

 **Recommendation:** Use continued pretraining for cost-effective specialization when targeting multiple tasks



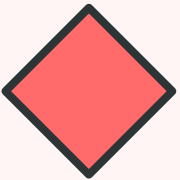
08

Conclusions & Future Work

Synthesizing key insights and outlining directions for future research



Key Takeaways



1

Specialist Efficiency

Small specialist LM can outperform larger generalist LM on targeted domain while having lower inference cost

2

CRISP Enables Specialists

CRISP enables pretraining specialist LMs when little specialization data is available - relaxing the data bottleneck

3

Systematic Superiority

Clustered importance sampling systematically outperforms alternative methods across multiple domains and tasks

4

Strong Performance

Pretraining with CRISP provides strong models both for language modeling and question answering tasks

5

Cost Reduction Strategies

Training cost can be lowered through:

Shorter Training

Still effective with fewer steps

Continued Pretraining

Share generic phase cost

Multitask Settings

Single model for multiple tasks

30×

Training Speedup

6

Tasks Evaluated

3

Model Sizes

16.7M

Max Clusters

Practical Recommendations



Use SBERT Embeddings

SBERT provides better clustering quality than LSI, leading to improved task-adaptive performance



Optimal Cluster Count

Around 260K clusters provides best balance for most applications – not too coarse, not too fine



Sufficient Generalist Data

Ensure sufficient generalist data to avoid excessive repetition when sampling from few clusters



Small Specialist Datasets

For small specialist datasets, consider augmentation or task grouping to improve cluster frequency estimation



Continued Pretraining

Continued pretraining offers cost-effective alternative to full CRISP training – share generic phase



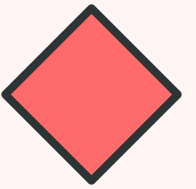
Multitask Weighting

Multitask weighting works well for related tasks – single model can serve multiple purposes



Remember: CRISP is modality-agnostic – can extend to vision, audio, and other modalities

Future Research Directions



Other Modalities

Extend CRISP to vision, audio, and multimodal settings – the method is modality-agnostic

High Impact



Better Estimation

Improve cluster frequency estimation for small specialist datasets through augmentation or task grouping

Addresses Limitation



Adaptive Clusters

Explore adaptive cluster count selection based on dataset characteristics and task complexity

Automation



Hierarchical IS

Investigate hierarchical importance sampling leveraging the tree structure for more nuanced reweighting

Method Extension



Quality + Domain

Combine CRISP with data quality filtering methods for joint optimization of quality and domain fit

Integration



Online Clustering

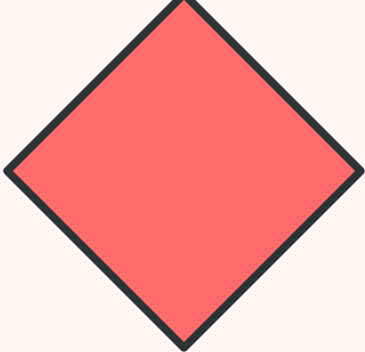
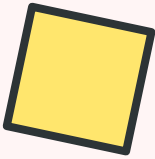
Study online clustering during training to adapt clusters as model learns and data distribution shifts

Dynamic Adaptation



Exciting Opportunities Ahead

CRISP opens new possibilities for efficient, task-adaptive model training across modalities and applications



Thank You

Questions & Discussion

Paper:

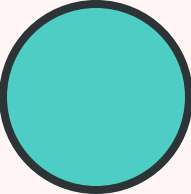
Task-Adaptive Pretrained Language Models via Clustered Importance Sampling

Authors:

David Grangier, Simin Fan, Skyler Seto, Pierre Abhin (Apple)

Venue:

ICLR 2025



Data Selection